

Exercises

Problems 1–10 are Core Exercises and contain the essential work for this chapter. Problems 11–20 are Challenge Problems at approximately AMC 8 and MATHCOUNTS level. They are optional for students whose main goal is a solid mathematical foundation, but are strongly recommended for students preparing for competitions. All exercises below have been newly written for this book.

Part A. Core Exercises

1. A 30-inch ribbon is cut into two pieces in the ratio $2 : 3$. Find the length of each piece.
2. A point P lies between 4 and 19 on a number line. If the ratio of $P - 4$ to $19 - P$ is $2 : 3$, find the coordinate of P .
3. A drink contains 6 ounces of syrup and 18 ounces of water. How much water must evaporate so that syrup becomes 40% of the remaining mixture?
4. Nina walks at 3 miles per hour. Forty minutes later, Omar starts from the same point at 5 miles per hour. How many hours after Omar starts will he catch Nina?
5. Two cyclists start 126 miles apart and ride toward each other at 16 and 20 miles per hour. How long does it take them to meet?
6. A boat travels at 12 miles per hour in still water. The current is 3 miles per hour. How long does it take the boat to travel 30 miles upstream?
7. One pipe fills a tank in 10 hours and another fills it in 15 hours. How long will they take if both are used together?
8. Machine A completes a job in 9 hours. Machines A and B together complete the job in 6 hours. How long would Machine B take alone?
9. Find the smaller angle between the hands of a clock at 4:20.
10. A recipe uses flour, oats, and nuts in the ratio $5 : 3 : 2$. If 4 pounds of the mixture are prepared, how many pounds are oats?

Part B. Challenge Problems

11. A point P divides the interval from a to b internally in the ratio $2 : 5$. If $P = 18$ and $a = 8$, find b .
12. A 20% salt solution has 5 ounces of pure water evaporated from it. The resulting solution is 25% salt. How many ounces were in the original solution?
13. Two runners start at the same point on a circular track and run in the same direction. One runs at 7 meters per second and the other at 5 meters per second. The track is 300 meters long. How many seconds after the start will the faster runner first lap the slower runner?

14. A train 180 meters long passes a pole in 12 seconds. At the same constant speed, how many seconds will it take the entire train to pass through a tunnel 420 meters long?
15. A plane flies 600 miles with a steady wind and returns the same 600 miles against the wind. Its speed in still air is 250 miles per hour, and the total flying time is 5 hours. Find the wind speed.
16. Pipe A fills a tank in 6 hours, Pipe B fills it in 8 hours, and a drain can empty a full tank in 24 hours. If all three are open, how long will it take to fill an empty tank?
17. At what time between 3:00 and 4:00 do the hands of a clock first form a right angle? Give the number of minutes after 3:00 as a fraction.
18. A car travels the first half of a 120-mile trip at 40 miles per hour and the second half at 60 miles per hour. What is its average speed for the entire trip?
19. Worker A can complete a job in 12 hours. A and B work together for 3 hours, after which A leaves. B finishes the remaining job in 6 more hours. How many hours would B need to complete the entire job alone?
20. At noon, a clock is correct. After that, its minute hand moves at the normal rate, but its hour hand moves 20% faster than normal. When the clock face first appears to show the hands exactly opposite each other, how many real minutes have passed since noon?

Part C. Beyond the Chapter

These questions are optional. They are not mainly about getting a numerical answer. Their purpose is to practice asking questions, deriving formulas, and summarizing ideas.

21. Derive the general section formula for a point x that divides the interval from a to b in the ratio $p : q$. Then explain in words why the coefficient of b is p , not q .
22. A slower traveler moves at speed v and has a head start of h hours. A faster traveler moves at speed $u > v$. Derive the catch-up formula

$$t = \frac{vh}{u - v}.$$

Then describe what happens to t as u gets closer and closer to v .

23. Derive a formula for the combined time of two workers who can complete a job separately in a and b hours. Your final formula should contain no fractions inside fractions.
24. Write a five-sentence summary of this chapter without using any of the numbers appearing in the examples. Your summary should explain at least one connection among ratios, motion, work, and clocks. Then create one new word problem whose solution uses two ideas from the chapter.

Answer Key

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|----------------------|---------------------|
| 1. 12, 18 | 2. 10 |
| 3. 9 oz | 4. 1 hour |
| 5. $7/2$ hours | 6. $10/3$ hours |
| 7. 6 hours | 8. 18 hours |
| 9. 10 degrees | 10. $6/5$ lb |
| 11. 43 | 12. 25 oz |
| 13. 150 seconds | 14. 40 seconds |
| 15. 50 mph | 16. 4 hours |
| 17. $360/11$ minutes | 18. 48 mph |
| 19. 12 hours | 20. $100/3$ minutes |

For Problem 21,

$$x = \frac{qa + pb}{p + q}.$$

For Problem 22,

$$t = \frac{vh}{u - v}.$$

For Problem 23, if the combined time is T , then

$$\frac{1}{T} = \frac{1}{a} + \frac{1}{b}, \quad T = \frac{ab}{a + b}.$$

Problem 24 is open-ended; a strong response should identify relationships, rates, units, and the habit of generalizing a solved problem.